QUESTION 1

What does the CPU fundamentally consist of?

Answer is Correct **Score: 1 / 1**

Options

A. ALU, memory blocks, and IO pins

Correct

B. ALU, registers, and control logic

C. Registers, IO drivers, and WiFi

D. Caches and wireless modules

FEEDBACK

Solution: A CPU consists of an Arithmetic Logic Unit, registers, and control logic.

Link: https://youtu.be/A4a18iJ48oE?feature=shared&t=81

QUESTION 2

How does a CPU communicate with memory or IO devices?

Not Attempted **Score: 0 / 1**

Options

A. Using UART

B. Through address decoding

Expected

C. Using a set of address and data lines called a bus

D. Only through system calls

FEEDBACK

Solution: The CPU uses a bus composed of address, data, and control lines to read/write to memory or IO.

Link: https://youtu.be/A4a18iJ48oE?feature=shared&t=135

QUESTION 3

What is the purpose of a memory-mapped IO block?

Not Attempted **Score: 0 / 1**

Options

A. To save power

B. To store program instructions

Expected

C. To behave like memory for the CPU but perform IO functions

D. To filter voltage noise

FEEDBACK

Solution: Memory-mapped IO blocks respond like memory when accessed by the CPU, but perform IO actions.

Link: https://youtu.be/A4a18iJ48oE?feature=shared&t=474

QUESTION 4

What is a GPIO in the context of a microcontroller?

Not Attempted **Score: 0 / 1**

Options

A. An analog-only pin

Expected

B. A general-purpose input or output pin

C. A read-only pin

D. A USB communication line

FEEDBACK

Solution: GPIOs are pins that can be used either as input or output and can be configured as needed.

Link: https://youtu.be/A4a18iJ48oE?feature=shared&t=638

QUESTION 5

What does writing ‘1’ or ‘0’ to a GPIO register do?

Not Attempted **Score: 0 / 1**

Options

A. Sets memory flags

B. Changes clock frequency

Expected

C. Alters pin voltage to high or ground respectively

D. Enables UART

FEEDBACK

Solution: Writing to GPIO registers sets the physical voltage level on the corresponding microcontroller pin.

Link: https://youtu.be/A4a18iJ48oE?feature=shared&t=694

QUESTION 6

What happens if both CPU and external world try to drive a GPIO at the same time?

Not Attempted **Score: 0 / 1**

Options

A. The pin automatically disables

Expected

B. A short circuit can occur

C. Nothing happens

D. A new interrupt is generated

FEEDBACK

Solution: Simultaneous driving with conflicting values can lead to a short circuit and potential damage.

Link: https://youtu.be/A4a18iJ48oE?feature=shared&t=905

QUESTION 7

Why are all GPIOs set as input by default during power-up?

Not Attempted **Score: 0 / 1**

Options

A. To allow high-speed toggling

B. To support analog signals

Expected

C. To prevent short circuits by avoiding output driving before configuration

D. To enable serial communication

FEEDBACK

Solution: GPIOs default to input to avoid unsafe driving conditions that can damage the circuit.

Link: https://youtu.be/A4a18iJ48oE?feature=shared&t=1029

QUESTION 8

What does it mean when a GPIO pin is said to be “floating”?

Not Attempted **Score: 0 / 1**

Options

A. It's a wireless pin

B. It’s unpowered

Expected

C. It is disconnected and has undefined voltage

D. It is used for analog only

FEEDBACK

Solution: A floating pin is one that is not connected to a defined voltage, making its state unpredictable.

Link: https://youtu.be/A4a18iJ48oE?feature=shared&t=1333

QUESTION 9

What is the role of pull-up and pull-down resistors in GPIO?

Not Attempted **Score: 0 / 1**

Options

A. They reduce heat

B. They change the clock rate

Expected

C. They ensure a known default voltage when the pin is unconnected

D. They filter analog signals

FEEDBACK

Solution: Pull-up/down resistors help prevent floating by holding the voltage at a known logic level.

Link: https://youtu.be/A4a18iJ48oE?feature=shared&t=1471

QUESTION 10

What kind of external devices often use open-drain configurations?

Not Attempted **Score: 0 / 1**

Options

A. Capacitors

B. Push-buttons

Expected

C. Sensors and chips that share common data lines

D. LED drivers

FEEDBACK

Solution: Open-drain devices like sensors allow multiple components to share a line without conflict.

Link: https://youtu.be/A4a18iJ48oE?feature=shared&t=1591

QUESTION 11

How does Arduino IDE (and similar environments) manage GPIO configurations easily?

Not Attempted **Score: 0 / 1**

Options

A. By using physical jumpers

B. Through low-level register toggling

Expected

C. Through Hardware Abstraction Layer (HAL) functions

D. By auto-detection logic

FEEDBACK

Solution: HAL provides functions that abstract low-level register operations for GPIO configuration.

Link: https://youtu.be/A4a18iJ48oE?feature=shared&t=1800

QUESTION 12

Why is HAL useful in embedded systems programming?

Not Attempted **Score: 0 / 1**

Options

A. It controls the CPU clock

B. It locks GPIOs to specific pins

Expected

C. It allows reusable and portable GPIO code across microcontroller families

D. It avoids use of interrupts

FEEDBACK

Solution: HAL abstracts hardware specifics, enabling code portability and reducing direct register manipulation.

Link: https://youtu.be/A4a18iJ48oE?feature=shared&t=1892